The University of Zambia

in association with

ZCAS University

BFS/BEF322/BAF422 TREASURY MANAGEMENT

FINAL EXAMINATION

1st DECEMBER 2021

12:30-15:30

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This question paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

QUESTION 1

Mr Bwalya is the proprietor of Bwalya and Sons, a chain of grocery shops operating from the local townships. Bwalya and Sons has been in operation for the past three years and has seen exponential growth in terms of number of shops and clientele. Mr Bwalya's success has been based on a philosophy of 'never turning a customer away' due to lack of stock or lack of money. To this end he always ensures that his shops are always fully stocked and he grants his customers generous credit facilities. As a result, his shops are the destination of choice for most customers in the townships. Mr Bwalya has plans to expand to more townships in the near future. Bwalya and Sons has been enjoying an overdraft facility from the bank, which has been renewed on two occasions without a problem. However, the last discussions with the Bank Manager to renew the overdraft did not go as well as Mr Bwalya had hoped. Mr Bwalya had requested his bank to increase the limit of the overdraft as he felt that the current limit was inadequate to meet his company's growing working capital needs. The Bank manager explained to Mr Bwalya that the Bank was not ready to increase the company's overdraft limit as they felt that the company was overtrading. Mr Bwalya cannot understand the bank's stand as according to his understanding his business was doing very well as evidenced by the increasing sales and profits that it was generating. Below are some selected figures from the company's financial statements.

Item	**2018**	**2019**	**2020**
	ZMK'000	ZMK'000	ZMK'000
Sales	2,400	5,600	8,400
Cost of Goods Sold	1,200	4,200	5,300
Cash	1,000	0	0
Overdraft	1,000	1,500	2,500
Stock	500	2,500	3,500
Debtors	400	1,000	2,500
Creditors	200	1,500	2,500

(i) From the figures above calculate the length of the cash cycle for the three years and comment on the results (confine your comments on the figures provided above only). **(15 marks)**

(ii) Comment on the reasons the bank might be reluctant to increase the overdraft limit. **(5 marks)**

(iii)Mr Bwalya is considering raising funding through his receivables. He has heard of factoring and invoice discounting as methods by which he can finance the company's working capital needs. He is not sure which method he should use.

a) Compare and contrast between the two methods of financing i.e. factoring and invoice discounting. **(10 marks)**

b) State the advantages and disadvantages of using factoring and invoice discounting. **(10 marks)**

(Total: 40 marks)

SECTION B: **Attempt any THREE questions in this section**

QUESTION 2

ABC Ltd. uses EOQ logic to determine the order quantity for its various components and is planning its orders. The Annual consumption is 80,000 units, Cost to place one order is ZMK 1,200, Storage cost per unit is ZMK 50.

(i) Calculate: EOQ, number of orders per year, ordering cost, storage cost and total cost of inventory. **(10 marks)**

(ii) ABC is considering switching to a JIT system of managing its inventory. Critically analyse this system and identify the merits and demerits of implementing the JIT system. **(10 marks)**

(Total: 20 marks)

QUESTION 3

"A well-designed and well-functioning payment system positively contributes to the financial stability of the country and to the well-functioning of the country's economy. The role of the Bank of Zambia is to ensure that Zambian payment systems are not only safe, but secure, reliable and efficient." (Extract from Bank of Zambia)

(i) Identify and describe the four (4) components of the payment system in Zambia. **(8 marks)**

(ii) Discuss how an efficient payment system contributes to the country's economy and the risks posed by an inefficient payment system. **(6 marks)**

(iii) The Bank of Zambia has also put in place measures to try and improve the payments and settlement systems in Zambia. Write short notes on the following payment and settlement systems and for each system suggest any two advantages of the system.

(a) Zambia Interbank Payment and Settlement System (ZIPSS) also known as the Real-time Gross Settlement (RTGS) System. **(2 marks)**

(b) Electronic Funds Transfer (EFT). **(2 marks)**

(c) The Cheque Image Clearing (CIC) system. **(2 marks)**

(Total: 20 marks)

QUESTION 4

ABC Company can borrow from its bank at a fixed rate of 4% or at a variable rate of LIBOR+1. XYZ on the other hand can borrow from its bank at a variable rate of LIBOR+2 or at a fixed rate of 6%. ABC Company has got a comparative advantage borrowing fixed but would like to borrow at a floating rate. XYZ would like to borrow at the floating rate. ABC Company and XYZ Company enter into one-year interest rate swap with a nominal value of $1 million. ABC offers XYZ a fixed annual rate of 5% in exchange for a rate of LIBOR plus 1%, since both parties believe that LIBOR will be roughly 4%.

(i) Give reasons why the two companies would want to enter into this agreement? **(5 marks)**

(ii) Identify the risks to both parties of entering into this arrangement and explain how they can mitigate the risk. **(5 marks)**

(iii)With the aid of a diagram, illustrate the amount of interest that will be exchanged between the two parties under the agreement. **(10 mark)**

(Total: 20 marks)

QUESTION 5

Average Exchange rates: 24-11-2021

Currency	**Buying**	**Selling**
USD	17.6960	17.7460
GBP	23.6347	23.7068
EUR	19.8301	19.8932
ZAR	1.1145	1.1178

(i) Given that the yield on the 91-day treasury bill rate in Zambia is 14.000% and the yield on the 3-month treasury rate in the U.S.A is 1.66%, calculate the 90-day forward rate for USD/ZMK using the rate table above (Note: use the USD/ZMK mid-rate for spot). **(6 marks)**

(ii) Discuss the three types of foreign exchange exposure that a firm dealing in a foreign currency might be exposed to. **(6 marks)**

(iii) Suggest any two (2) business practices that a firm can use to manage its foreign currency risk that do not involve the use of financial derivative instruments. Discuss the limitations of the practices that you choose. **(8 marks)**

(Total: 20 marks)

END OF EXAMINATION

MARKING KEY

Question 1

(i)

	2018	2019	2020
Inventory days: Inventory/COGS x 365	500/1200 x 365 = 152	2500/4200 x 365 =217	3500/5300 x 365 =241
PLUS			
Debtor Days: Accounts Receivable/Sales x 365	400/2400x365 = 60	1000/5600 x365 = 65	2500/8400 x365 = 108
	212	282	349
MINUS			
Creditor Days: Accounts Payable/COGS x 365	200/1200 x 365 = 61	1500/4200 x 365 =130	2500/5300 x365 =172
COGS	151	152	177

- Mr Bwalya is facing a liquidity crisis evidenced by reduced cash and increasing need for an overdraft facility.
- The cause of the liquidity crisis could be as a result of his policy of holding more stock and granting generous credit to his customers.
- This has resulted into increasing inventory days, debtors days and creditor days.
- Increasing inventory days and debtor days means that the firm is less liquid as cash is tied up in stock and debtors.
- This in turn means that the firm takes longer to meet its own commitments to the creditors.
- The CCC has worsened over the years from 151 days (2018) to 177 days (2020)

(ii) Reasons on why the Bank might be reluctant to increase the overdraft:

- There is evidence of overtrading by the firm
- The liquidity position has worsened over the years
- The firm may need higher limits which may come at a higher cost.
- The business needs to work on its working capital management

(iii)

(a)

	Factoring	Invoice discounting
Comparison	• Method of raising short term funds • Usually unsecured	• Method of raising short term funds • Usually unsecured
Contrast	• Non-confidential facility • Factor involved in the administration of sales ledger • Financing may be without recourse • Credit risk may be transferred to the factor	• Confidential facility • Financier not involved in administration of sales ledger • Always with recourse to the client • Credit risk remains with the client

(b) Advantages:

- Flexibility – as sales increase with the corresponding demand for finance, so finance from this source increases
- Security- it allows the firm to pledge other assets as security for the finance
- Last resort- it may be the most cost-effective lender to a firm which has no assets to offer as security
- Responsibility- relieves management of the responsibility for the sales ledger and can probably perform credit checking better than the firm.

Disadvantages:

- Customer resistance
- Customer relationship
- Reputation
- Too reliant

Question 2

(i) Annual Demand = 80,000

Ordering cost = K1,200

Holding cost = K50 per unit

$$EOQ = \sqrt{\frac{2ac}{h}}$$

$$= \sqrt{\frac{2 \times 1{,}200 \times 80{,}000}{50}}$$

$$= \sqrt{\frac{192{,}000}{50}}$$

= 1960 units

Number of orders per year = 80,000/1960

= 41 orders

Ordering Cost = 41 x 1,200

= 49,200

Holding cost = 1960 x 50

= K98,000

Total Cost = 49,200+98,000

= K147,200

(ii)

- Just in time (JIT) is an inventory management system, used to manage the stock that is kept in storage.
- It involves receiving goods from suppliers as and when they are required, rather than carrying a large inventory at once.

- JIT purchasing means that materials or components are received from the supplier just in time to use them

ᘓ Objectives of JIT purchasing:

- Raw material stock reduction;
- Frequent deliveries with smaller orders from a smaller number of suppliers;
- Long term contracts with suppliers
- Quality assurance, with supplier becoming responsible for inspection of goods supplied

Merits of JIT system:

- Reduced inventory
- Savings in storage space required
- Increased customer satisfaction – elimination of waste leads to better quality product
- Less likelihood of stock perishing, obsolete stock, pilferage etc

Demerits of JIT system:

- Little room for mistakes as minimal stock is kept
- Production reliant on suppliers
- No spare finished products available to respond to unexpected orders
- risk running out of stock

Question 3

(i) Four components of the payment system in Zambia:

- A payment system includes
- the participants (institutions)
- and the users (customers/clients),
- the rules and regulations that guide its operation and
- the standards and technologies on which the system operates.

(ii) How an efficient payment system contributes to economic growth:

A well-designed and well-functioning payment positively contributes to the financial stability of the country and to the well-functioning of the country's economy

- Reduces settlement and payment risk in the economy
- Reduces the cost of doing business in the country
- Velocity of money is increased in the economy resulting in more business transactions.

(iii) ZIPSS

- ZIPSS is an electronic payment system in which processing of transactions for settlement takes place continuously on a transaction basis in real time
- ZIPSS was developed to improve the management of payment risks related to high value payments.
- It provides a more secure platform for making both large value payments
- It also provides quicker final and irrevocable payments

Benefits of using ZIPSS:

- ZIPSS carries negligible credit and settlement risk because it is based on pre-funding of payments before a payment instruction is issued

- ZIPSS provides banks with the facility to monitor their settlement balances throughout the day as each transaction takes place in real time

- Due to the continuous nature of transfers, cleared funds are available to the recipient with no significant time lag
- ZIPSS/RTGS facilitates the linkage to other transfer systems such as money and capital markets
- ZIPSS/RTGS enhances treasury management for both banks and bank customers
- Payments that are made on ZIPSS are final and irrevocable
- ZIPSS being a real-time settlement system means that payments take effect on appropriate accounts immediately
- No minimum or maximum amount is imposed on ZIPSS and therefore customers are able to process both low and large value payments on ZIPSS

Electronic Funds Transfer (EFT)

- The Electronic Funds Transfer (EFT) system is a simple, safe and speedy way to make and collect payments by electronic transfer
- Safe and secure transfer of funds from one account to another

Benefits of EFT

- Some of the benefits of the EFT system include;
- Credit to the customer is given within the same day
- Allows a transfer of higher value (has limit than cheques)
- It is safe and secure and guaranteed through registered commercial bank
- High availability since the service is provided by almost all commercial banks

The Cheque Image Clearing System

- The Cheque Image Clearing (CIC) system is a cheque processing and clearing system and allows customers belonging to one bank to write cheques belonging customers of another bank.
- Similar to Electronic Funds Transfer (EFT) system, the CIC system is operated by the Zambia Electronic Clearing House Ltd (ZECHL).
- Cheque truncation system (CTS) is a system that involves the process of clearing cheques using cheque images between banks as opposed to using physical cheques presented for payment in a bank by individuals or corporate customers.

Question 4

(i) Reasons why the two companies may want to enter into a swap arrangement:

• The parties may enjoy a comparative advantage borrowing in one type of interest rate as opposed to the other.

• The parties may want to manage their interest rate exposure by swapping their interest rate commitments.

• The parties may want to profit from the arrangement by entering into a speculative arrangement

(ii) Risks to the parties of entering into a swap agreement:

Interest rate risk

• The actual interest rate movements may not match the expectations of the parties

• For example, the counter party receiving a fixed rate payment stream loses if interest rates rise and vice versa

Counter party credit risk

• There is a risk that the counter party will default on their commitments.

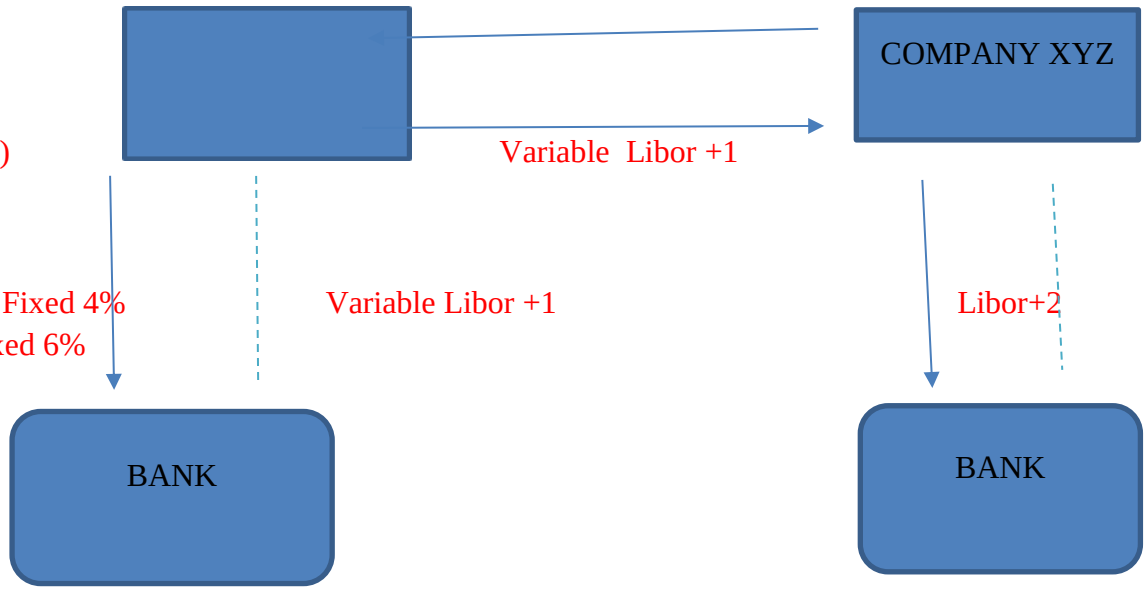
Mitigation

• The parties can conduct the swap transaction through a bank intermediary to reduce the credit risk

• The parties may be required to pledge collateral as part of the transaction.

• The parties are required to sign swap agreements

(iii)



Company ABC

Pays fixed to bank = (4%)

Receives fixed from XYZ= 5%

Pays variable to XYZ = (5)%

Company XYZ

Pays variable to bank = (6%)

Receives variable from ABC = 5%

Pays fixed to ABC = (5%)

Interest per annum to be exchanged between the parties = 5% x $1,000,000 = $50,000

Question 5

(i) $TB_z = 14.000\%$

$TB_{us} = 1.66\%$

Mid rate USD/ZMK = 17.6960+17.7460/2 = 17.7210

Term = 90 days

$$F_{USD/ZMK} = 17.7210 \times \frac{(1+0.14/4)}{(1+0.0166/4)}$$

$$= 17.7210 \times \frac{1.035}{1.00415}$$

$$= 17.7210 \times 1.0307$$

$$= 18.2654$$

(ii) Types of FX exposure:

Economic risk

- Economic exposure directly impacts the value of a firm.
- That means, the value of the firm is influenced by the foreign exchange.
- The value of a firm is the function of operating cash flows and the assets it possesses.
- The economic exposure can have bearings on assets as well as operating cash flows. Identification and measuring of this exposure is a difficult task.
- Although, the asset exposure is still measurable and visible in books but the operating exposure has links to various factors such as competitiveness, entry barriers, etc which are quite subjective and interpretation of different experts may be different.

Transaction risk

- this exposure pertains to the exposure due to an actual transaction taking place in business involving foreign currency.

- If you have bought goods from a foreign country and payables are in foreign currency to be paid after 3 months, you may end up paying much higher on the due date as currency value may increase.
- This will increase your purchase price and therefore the overall costing of the product compelling the profit percentage to go down or even convert to lose.
- Transaction exposure normally occurs due to foreign currency debtors of sale, payment for imported goods or services, receipt / payment of dividend,

Translation risk

- This exposure is also well known as accounting exposure.
- It is because the exposure is due to the translation of books of accounts into the home currency.
- Translation activity is carried out on account of reporting the books to the shareholders or legal bodies.
- Gains or losses arising out of translation exposure do not have more meaning over and above the reporting requirements.
- Such exposure can even get reversed in the next year translation if currency market moves in the favorable direction.
- This kind of exposure does not require too much of management attention.

(iii) Business Practice that a firm may use to manage its FX exposure:

1. Insist on home currency payment
2. Currency surcharges
3. Get paid on time
4. Foreign currency loans
5. Sourcing Changes
6. Foreign Currency Accounts
7. Unilateral, Bilateral and Multilateral Netting Arrangements